

Clinton Judy

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SUMMARY

I build production AI products end to end. At Zipline I created Zippy, the company's flagship AI assistant, and led it for three years: 50+ retail brands, 1,000,000+ conversations, two re-architectures under live traffic, and agentic tool use over MCP. I shipped contextual retrieval to paying customers about fifteen months before Anthropic published the technique. Underneath is 15+ years as a Rails engineer and SRE at scale. Today I'm building my own agent harness from scratch.

WORK EXPERIENCE

Senior Web Engineer

May 2019 — Sep 2026

Zipline

Remote

AI Systems (2023–2026)

- **Created “Zippy”, Zipline’s production RAG assistant and flagship AI feature:** found the approach in research, pitched it, built the prototype in two weeks, named it, and shipped it. Live with **50+ retail brands** across **1,000,000+ conversations** and **2,000,000+ pages** indexed; the product’s primary AI differentiator for three years.
- Arrived independently at contextual retrieval about fifteen months before it was published as a technique: built ingestion that decomposes documents into self-contained **atomic facts**, the same core insight Anthropic later published as Contextual Retrieval.
- **Re-architected information retrieval engine, twice.** Led the migration from pgvector to Elasticsearch. Then designed and shipped vision-model document understanding, so Zippy could answer questions about diagrams, planograms, and tables.
- **Shipped agentic tool use to production**, including MCP (Model Context Protocol) integration that lets Zippy query live systems for real-time inventory, scheduling, and policy answers. Built the MCP server model, client, and admin interface, and multi-step tool calling via the OpenAI Responses API.
- **Made retrieval quality measurable:** built Zippy’s first LLM-as-judge evaluation system and multiple evaluation data sets that informed later architecture decisions, including a scenario set generated from three years of real customer bug reports, so migrations were decided on measured pass rates instead of opinion.
- Engineered fast, access-controlled retrieval at enterprise scale: vector search combined with permission-predicate filtering, working around HNSW’s degradation under predicate constraints so results stay fast and correctly scoped per user, team, and role.
- Built the full conversational front end with Hotwire (Turbo + Stimulus): token-streaming responses, multi-turn state, source citations, query refinement, and inline feedback capture.
- Shipped a multilingual retrieval fix over a weekend to unblock a customer’s pre-recorded national television segment, adding automatic query translation so non-English questions retrieved correctly.

Architecture & Technical Leadership (2025–2026)

- Settled architecture arguments with measured evidence: ran a corpus study on real customer documents to prove which extraction failures actually occurred, won committee approval for the result over the chair’s preferred alternative, and had it in production fifteen working days later.
- Set the numeric go/no-go criteria my own projects were judged against, before results came in.
- Challenged a projected \$100K/yr infrastructure spend by pricing it against the existing pipeline’s invoiced cost, then ran the vendor review that surfaced a 32x cheaper path.
- Served as the company’s AI technical authority: standing member of the AI Steering Committee, author of the RFCs setting AI architecture direction, escalation point for AI questions from engineering, product, sales, and customers; fronted the platform’s AI story in enterprise security and responsible-AI reviews, sales RFPs, and vendor negotiations.
- Acted as team lead for the AI/search team: cycle planning, technical direction, and distributing architecture work across the team rather than holding it.

Product & Platform Engineering (2019–2023)

- Built Assessment tools end to end: historic results, auditee search, scoring and sorting, and the complete auditor workflow. **116M+ assessment questions answered.**
- Built Report Builder (custom columns, filters, role-based access), Learning Insights learner reports, and discussion threading with notifications and mentions.
- Built platform capabilities: REST Hooks for Zapier, a video transcoding pipeline on AWS MediaConvert with HLS streaming, and MegaSeed, an internal large-scale realistic-data generator for performance testing.
- Migrated legacy reporting to BigQuery as part of a small team, matching edge cases exactly so historical numbers stayed correct.
- Kept the application fast as Zipline grew 10x in active users, through Rails and SQL optimization, caching, and automated performance testing in CI.

Software Engineer

Oct 2018 — Apr 2019

Flatiron School

New York, NY

- Consolidated logs, metrics, and traces across a large microservice architecture, increasing visibility and reducing MTTR; helped migrate services from Digital Ocean to AWS.

Senior Site Reliability Engineer

Dec 2016 — Oct 2018

AccuWeather

State College, PA

- Built the CI/CD pipeline and a monitoring system tracking hundreds of statistics across tens of millions of daily requests; automated deploys across public-facing AccuWeather properties. Contributed to Ansible and Terraform.

Senior Web Developer

Feb 2016 — Oct 2016

EFC Systems

State College, PA

- Built data-visualization dashboards (JavaScript + D3) and front-end tooling for precision agriculture; overhauled authentication to modern encryption standards.

Senior Rails Engineer

Nov 2013 — Jan 2016

Agrian Inc.

State College, PA

- Built service-oriented Rails applications and large OO frameworks for diverse raw data formats, optimized for big-data scale (100M rows in the first month).

Open Source Lead

2012 — 2013

Fractured Atlas

New York, NY

- Turned Artful.ly, a CRM and online ticketing platform, into an open-source Rails engine; built discount, search, and customer-segmentation features and improved sign-up conversion.

Founder & Community Organizer

2010 — 2019

The Make Space / State College Ruby Group

State College, PA

- Founded The Make Space, a State College hackerspace, and ran the State College Ruby Group for several years, giving a recurring series of technical talks; consulted local startups on architecture, UX, and object-oriented design.

Web Developer

2009 — 2011

West Arete

State College, PA

- Designed, built, and launched data-driven Rails web applications for clients including MIT, Penn State, the DOT, and the Marine Biological Laboratory.

SELECTED PROJECTS

Kormos: a from-scratch agent harness (private, in daily production use)

Mar 2026 — Present

Independent

- Designed and built a complete LLM agent runtime in Rails 8, provider-agnostic across Anthropic, OpenAI, and local models: tool dispatch with 17 built-in tools, an MCP client layer with OAuth Dynamic Client Registration and per-provider token lifecycle, and 25 composable context enrichers that assemble each turn's prompt.
- Owns the hard parts of a harness rather than delegating them to an SDK: prompt-cache breakpoint placement, bounded image lifetimes so vision blocks stay out of cached prefixes, durable turns that survive process restarts and recover orphaned tool calls, and metered subscription pools with automatic failover.
- Runs agents across Telegram, web, SMS, and live phone calls (Twilio media streams, streaming STT, and cloned-voice TTS with barge-in), plus scheduled autonomy with a firing ledger and inbound webhooks, so an agent can act unprompted and be audited afterward.
- Six months in: 740+ commits, 570+ merged PRs, 479 test files, and a second independent operator running it on macOS whose bug reports have driven fixes within the day. Companion systems: Plektos (episodic memory with salience-weighted recall) and Thymos (an emotional-signal substrate that gives agents a basis to disagree).
- Built with a staged coding-agent pipeline I run across parallel worktrees, with a human checkpoint at design forks and dispatched architectural review before merge; the pipeline and its process skills are published at github.com/judy/skills.

SKILLS

- **AI/ML:** Retrieval-Augmented Generation (RAG), contextual retrieval, Large Language Models (LLM), LLM-as-judge evaluation, embeddings, hybrid vector + lexical search, agentic systems, tool use, MCP (Model Context Protocol) client and server design, prompt caching, multi-agent orchestration, episodic memory / consolidation
- **Languages & Frameworks:** Ruby on Rails (15+ years), JavaScript (15+ years), TypeScript (working), Python, Hotwire (Turbo + Stimulus), Svelte, React
- **Data:** PostgreSQL, pgvector, Elasticsearch (incl. Painless scripting), SQLite, Redis, BigQuery, Sidekiq, Solid Queue, Action Cable
- **Infrastructure:** Docker, Ansible, AWS, Terraform, CI/CD, systemd and launchd deployments, Twilio voice and SMS
- **Other:** D3, video transcoding (HLS), open-source contributions (PDF parsing, Ruby gems, Paseo)

EDUCATION

Lock Haven University of Pennsylvania

Coursework in Computer Science, Graphic & Online Design

Lock Haven, PA

Aug 2004 — May 2007